gpeyt1-34 · Collapsible soil settlement

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A 9 ft thick layer of dry loess will support a spread-footing warehouse. A double-oedometer test on a sample from mid-layer (initial e0 = 0.98) shows: under the design stress (dry) e = 0.94; after inundation at that same stress e = 0.83. Natural w = 6%, γd = 86 pcf. The owner asks how much additional settlement occurs if lawn irrigation wets the layer after construction.

Most nearly, the collapse settlement caused by wetting is:

A. 8.2 in.
B. 11.9 in.
C. 6.0 in.
D. 0.5 in.

Given equations / data

> $\Delta H = H\,\Delta e/(1 + e0)$

Worked solution

1. Δe (collapse) = 0.94 − 0.83 = 0.11
2. $\Delta H = 9 \times 0.11/(1 + 0.98) = 0.500$ ft = 6.0 in.

Final answer

6.0 in.

Why the other values are wrong

- **8.2 in.** — included the dry load-compression as collapse
- **11.9 in.** — forgot to divide by (1 + e0)
- **0.5 in.** — left the answer in feet labeled inches

References (Jayce's library)

- Vandevelde, ASCE PE Review — Foundation Engineering (S = HΔe/(1+e0) worked example); NCEES PE Civil Reference Handbook v2.1 §3.2 Consolidation (strain form)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU, https://www.youtube.com/watch?v=M9uBSVNmUgU (no creator values copied).

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